MIL-P-52235B(ME)
4 January 1985
SUPERSEDING
MIL-P-52235A(ME)
12 September 1978

MILITARY SPECIFICATION

PUMP, RECIPROCATING, HAND-DRIVEN, SINGLE-ACTING, BILGE

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hand-operated, single-acting, reciprocating, bilge pump.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2000 - Rubber Products in Automotive Applications.
D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J30 - Fuel and Oil Hoses.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pump shall consist of a cylinder, piston with rod and handle, and a discharge spout with discharge hose attachment.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of pumps from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish pumps conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The bilge pump shall be fabricated from compatible materials, inherently corrosion resistant or treated to be resistant to the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by a barrier such as a coat of epoxy primer conforming to MIL-P-23377 or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components or sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pump may be newly fabricated from recovered materials to the maximum extent practicable, provided the pump produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the pump.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

3.3.2 Rubber. Synthetic rubber for piston cup, strainer foot, and cylinder cap shall conform to ASTM D 2000, type 3BC710A14C12C20E14F17.

3.4 Construction.

3.4.1 Cylinder. The cylinder shall be heavy seamless copper alloy tubing. The cylinder ends shall be rolled to provide pressure tight sealing device.

3.4.2 Piston. The piston shall be a molded synthetic rubber cup, reinforced with a copper alloy plate, and affixed to a tapped and threaded copper alloy piston rod. The piston cup shall be removable from the piston rod without stretching.

3.4.3 Handle. The handle shall be D-shape, rigid plastic or copper alloy, and shall be tapped and threaded to accomodate the threads on the upper end of the piston rod.

3.4.4 Strainer foot. The strainer foot with integral valve shall be molded from synthetic rubber and shall be grooved to fit the rolled end of the cylinder.

3.4.5 Cylinder cap. The cylinder cap with integral discharge spout shall be synthetic rubber molded in one piece, and grooved to fit the rolled end of the cylinder.

3.4.5.1 Discharge spout. The discharge spout shall be copper alloy tubing, sized to receive and retain by friction, a one inch inside diameter (25.4 millimeter [mm]) synthetic rubber hose.

3.4.6 Discharge hose. A one inch (25.4 mm) inside diameter synthetic rubber hose 60 inches (1524 mm) in length shall be furnished with each pump. The hose shall conform to the requirements for type 30 R1 of SAE J30.

3.5 Performance. The pump shall be capable of pumping one US gallon (.0004 cubic meters [m^3]) of water in eight cycles.

3.6 Dimensions. Dimensions shall be as shown on figure 1 (see 6.4) or as specified herein.

3.7 Valve. The valve shall be clearly labeled with a double-ended arrow indicating open and closed position, i.e., open ⟷ closed.

3.8 Threads. All threads shall conform to FED-STD-H28.

3.9 Weight. The pump, complete with discharge hose, shall not exceed 5 pounds (2.27 kilograms).

3.10 Identification marking. The pump shall be identified in accordance with MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

3.11 Workmanship. All parts, componets, and assemblies including molded parts shall be clean and free from sand, dirt, fins, pits, scale, and other harmful extraneous material. All metal edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The pump(s) shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The pump(s) shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, inspection level II.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 6.5 percent defective.

4.4.3 Test.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1. AQL shall be 4.0 percent defective.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:10Z
Check the source to verify that this is the current version before use.

4.5 Inspection procedure.

4.5.1 Examination. The pump shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of materials, material finishes or treatments.
105. Rubber not as specified.
106. Cylinder not as specified.
107. Piston not as specified.
108. Handle not as specified.
109. Strainer foot not as specified.
110. Cylinder cap not as specified.
111. Discharge spout not as specified.
112. Discharge hose not as specified.
113. Dimensions not as specified.
114. Valve not labeled as specified.
115. Threads not as specified.
116. Weight not as specified.
117. Identification marking missing, incomplete, or illegible.
118. Workmanship not as specified.

4.5.2 Test.

4.5.2.1 Performance test. The cylinder shall be considered full of water after two full cycles. One full cycle consists of one up and one down operation of the handle. The pump shall be worked for eight full cycles. The discharge water shall be collected in a container. Nonconformance to 3.5 shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

119. Materials, methods, or containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:10Z
Check the source to verify that this is the current version before use.

120. Box not closed and sealed as specified for level A.
121. Strapping not as specified for level A.
122. Gross weight or size exceeds the limitations of the applicable box specification for level A or B.
123. Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The pump shall be disassembled to provide minimum cube in accordance with the manufacturer's recommendations and then preserved in accordance with MIL-P-116, method III, using a close-fitting box conforming to PPP-B-636, W6c, style optional. The box shall be closed and sealed as specified for method V of the appendix to the box specification.

5.1.2 Commercial. The pump shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. The pumps, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. The pumps, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c, style optional, in quantities not to exceed the size and weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial. The pumps, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on the shipping container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The pump is intended for use with floating bridge pontoons and other military floating equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of first article pumps and number of pumps required.
c. Degree of preservation and degree of packing required (see 5.1 and 5.2).
d. When strapping shall be other than as specified (see 5.2.1).

6.3 First article. When a first article inspection is required, the item should be a preproduction pump. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article.

6.4 Information figures. Figure 1 shows a type of pump which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of another pump which conforms to this specification.

Custodian:
Army - ME

Review activity:
DLA - CS

Preparing activity:
Army - ME

Project 4320-A219

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:10Z
Check the source to verify that this is the current version before use.

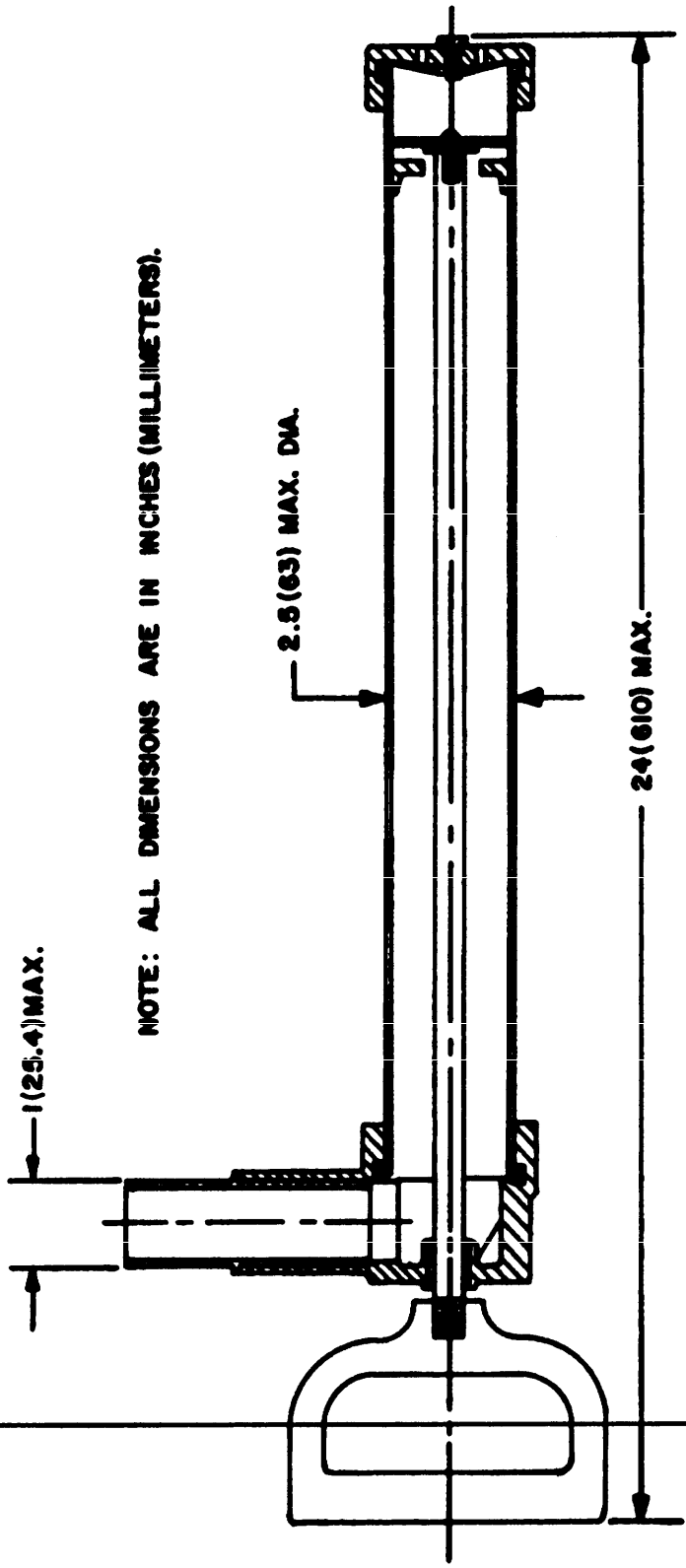


FIGURE 1. Schematic drawing for information only, configuration of pump is not mandatory.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Belvoir R&D Center
ATTN: STRBE-DS
Ft. Belvoir, VA 22060-5606

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-52235B	Pump, Reciprocating, Hand-Driven, Single-Acting, Bilge

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 2026-02-22T21:10Z
Check the source to verify that this is the current version before use.